

Two-way chemotaxis-based communication for biological nanonetworks

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ABSTRACT

This work proposes a new switching scheme between different chemotactic substances, as a way to affect bacterial movement and to enhance message delivery in biological nanonetworks. By switching between different chemotactic substances release plans, controlled by the stations, the accuracy of two-way communication in such networks is improved. The results indicate that the proposed conditional release of chemotactic molecules is effective for transmitting messages, enhancing reliability and reducing channel saturation.

CCS CONCEPTS

• **Networks** → **Network protocols**.

KEYWORDS

biological nanonetworks, chemotaxis, diffusion

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1 INTRODUCTION

We consider a biological nanonetwork that uses bacterial cells to disseminate information. The system consists of stations located inside a common fluidic medium. Bacteria concentration is used to encode messages under the method of on-off keying [2]. The bacteria reach their target by free or directional diffusing inside the medium dependent of the presence of chemotactic substance.

The behaviour of the bacteria is affected by the presence of chemotactic substances inside the medium, which can be used to direct them towards or away from a target. Bacteria move towards regions of higher concentration in case of chemoattractants or lower concentration in case of chemorepellents molecules in the medium. The movement of cells towards areas with higher chemotactic substance concentration is called positive chemotaxis and the substance is known as chemoattractant, while movement towards areas with lower substance concentration is called negative chemotaxis, and the substance is known as chemorepellent. The stations can, thus, affect the performance of the communications system by appropriately releasing such substances resulting in different diffusion patterns.

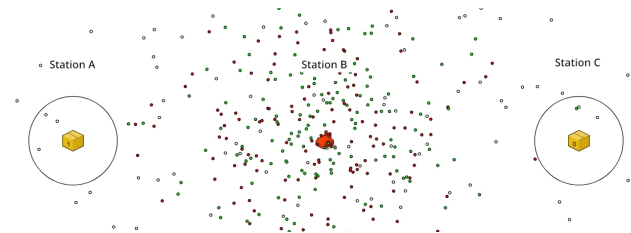


Figure 1: Network topology

Existing literature [4] examines the use of a single substance type, either chemoattractant or chemorepellent, to enhance the performance of the network. In our research, a switching protocol between the release of both types of

chemotactic substances is explored. Supplementary to the existing literature that studies systems being in an equilibrium state concerning the chemotactic substance, the proposed model lies on the boundaries of the chemotactic interface, capturing the dynamics of chemotactic substance interchange. As shown below, alternating between the release of chemoattractant and chemorepellent molecules results in higher message transmission reliability.

2 CHEMOTACTIC SWITCHING STRATEGY

2.1 Topology and simulation scenarios

The base topology consists of one central station and two stations placed symmetrically to it at a distance of $400\mu\text{m}$ (Fig. 1). Parameters affecting bacterial behaviour were based on [3]. Central station is present in order to have a point of reference for the concentration.

Multiple simulations were conducted to realise the effective chemotactic switching strategy. The main scenario involves station B sending a message in the form of releasing a number of information molecules (pulse of molecules) to the station A at 100s, followed by a response at 250s. We assume that station A is releasing chemoattractant starting from 0s as a result of previous messages sent to it. We expect the high chemoattractant concentration around station A to prevent bacteria in the reply pulse from reaching station B. Our chemotactic switching strategies under investigation include one where station A sends a pulse and immediately starts releasing chemorepellent (Fig. 2b) and another where station A does not release new information(signaling) bacteria, but emits chemorepellent to create a wave using the existing bacteria in the medium (Fig. 2c). To determine the effectiveness of the strategies, we compared them against a base scenario, in which station A sends out a pulse and immediately stops releasing chemoattractant molecules (Fig. 2a).

2.2 Simulation results

Fig. 2a shows that when station A ceases chemoattractant release and emits a pulse, station B notices an increase in bacterial concentration without being able to understand any peak. As a result, it cannot discern between a message pulse and a typical increase of bacterial concentration due to the added bacteria from previous transmissions. The effectiveness of the proposed chemorepellent policy is clearly visible in Fig. 2b, in which station B, at about 300s, notices an increase and decrease in bacteria concentration that resembles a pulse peak - interpreted as a message transmission from another station. The results of the second release strategy are presented in Fig. 2c, where station B is able to understand the message pulse originating from station A, even at lower

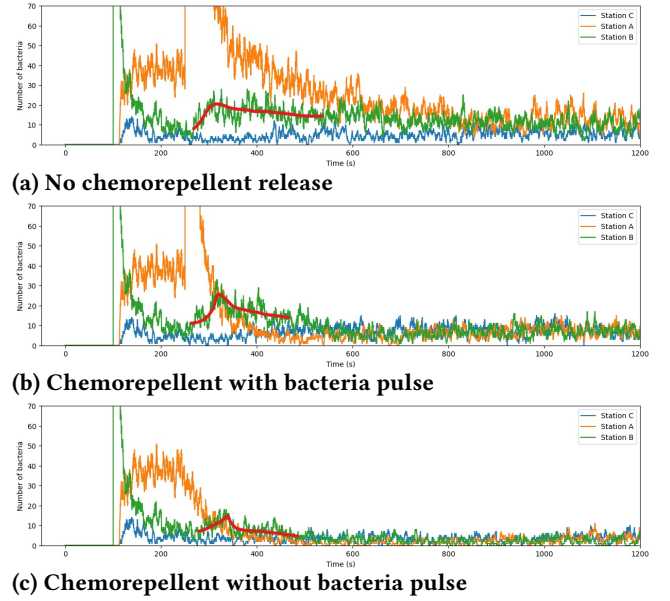


Figure 2: Bacteria concentration of stations

values of concentration of signaling molecules-bacteria. Bacteria concentration at station C is not affected by the different release policies applied in this work, as shown in Fig. 2a 2b, 2c.

3 CONCLUSION

This work proposed a release management protocol for chemotactic substances in biological nanonetworks with a focus on increasing transmission reliability. Simulation results prove the validity of the suggested approach. Our ongoing work entails different simulation scenarios in an effort to apply the scheme to multiple access biological nanonetworks as a future work, for which the present study lays the foundation.

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